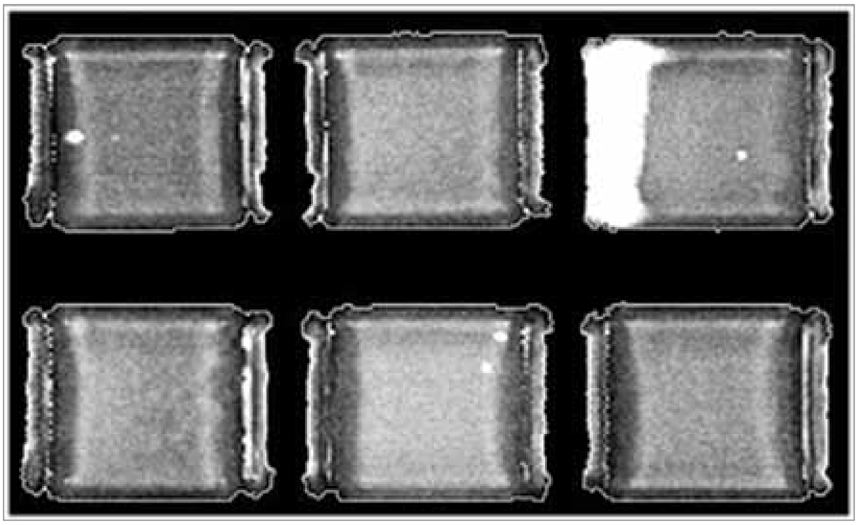

Fig. 2: Part of a tray of MLCCs; white features are defects

and delaminations. Voids are found in the dielectric and are spherical or almost any other shape, but are typically more three-dimensional than delaminations. Delaminations are typically flat air pockets between the dielectric and an electrode.

Under mechanical and electrical stresses of thermal cycling, both voids and delaminations are capable of expanding to cause a dielectric breakdown. Failure occurs when a conductive pathway is created through the dielectric between two electrode layers and becomes a leakage pathway or a short circuit.

It is beneficial, then, to find any voids or delaminations or other air gaps in an MLCC before it is mounted onto a board (or even after it is mounted, if necessary). Whether the AMI tool is imaging a single MLCC or whole trays of MLCCs, the imaging process follows this sequence:

- A pulse is launched by the transducer and travels 3mm or so through a water column to a precise x-y location on the surface of the MLCC.
- Part of the pulse is reflected by the interface between the water and the MLCC's surface back to the transducer, where its travel time in nanoseconds is recorded to provide a record of the precise distance between the transducer and the MLCC.
- The other portion of the pulse crosses the interface and travels into the ceramic-metal layers comprising the capacitor, where it is slightly reflected at each material interface.
- Within the capacitor, if the pulse strikes the top surface of an air-filled void or delamination, virtually all of the pulse is reflected to the transducer where its amplitude, arrival time, and sometimes other properties are measured. The pixel made from it will be bright white or bright red, the two colours typically used to indicate a reflection of maximum amplitude.
- If the pulse strikes no solid-to-air interface, it will continue on to the bottom of the capacitor and be reflected from there back to the transducer. From crossing so many minimally reflective interfaces, both going and returning, the pulse/echo will have lost energy.

Fig. 1 shows the acoustic image of an MLCC having a large internal defect. A pulse was launched into each of the thousands of x-y locations that make up the whole acoustic image, and a colour from the colour map at left was assigned to each location. Green pixels represent the dieletric-electrode stacks, while red is the packaging around them.

Solid white in the area at left represents echoes having the highest amplitude. This area is the interface between the electrode and the air in the delamination. None of the ultrasonic pulses traveled deeper in this area because ultrasound at these frequencies will not travel through air, even if the air is only a tiny fraction of a micron in thickness.

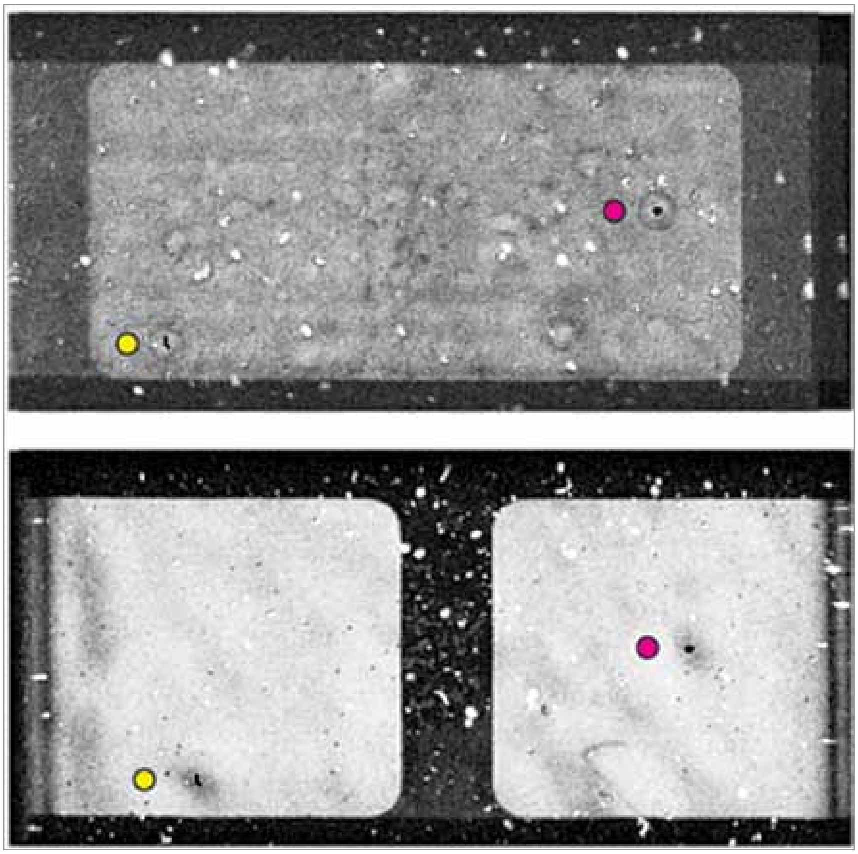

Fig. 3: Imaging of gates 3 and 4 in an MLCC